

PE6201 Assignment 2 – D6 Cost-to-Serve Analysis

Problem B: Outpatient Referral Coordination

Monthly volume: 4,000 referrals | Human fallback cost: US\$9.17 per failure

1. Section 4 – What it costs

Our cost-to-serve model follows the assignment's escalation-on-failure formulation. Layer 1 is measured token usage priced at each model's documented list rates; Layer 2 is $(1 - P) \times \text{US\$9.17}$, representing ten minutes of triage-nurse handling; Layer 3 was not measured because the prototype used local fixtures and no specified production infrastructure. At 4,000 referrals per month, Gemini 2.5 Flash Lite produced the lowest measured cost: US\$0.001131 in model cost plus US\$0.163750 in expected fallback, or US\$0.164881 per referral and US\$659.52 per month before Layer 3. GPT-5 Mini cost US\$1,328.32 monthly, Qwen US\$1,977.68, GPT-4o Mini US\$5,903.41, and Claude Haiku US\$19,666.56.

The sensitivity analysis shifted each measured success rate by ± 10 percentage points. Gemini ranged from US\$4,327.52 per month at 88.21% success to US\$4.52 at the 100% cap, and remained the least-cost model when the same shift was applied across all models. However, the model-specific ranges overlap, so the ranking should not be treated as uncertainty-free. Against GPT-5 Mini, Gemini's break-even success rate was 96.391%; its measured 98.214% exceeded this threshold by 1.823 percentage points.

The cost ledger measured all four design levers. Removing the model-visible *as_of* tool reduced the tool block from approximately 1,007 to 912 tokens, a 9.4% reduction. Parallel execution reduced mean turns from 4.625 to 3.625 and input tokens from 1,043,400 to 749,400 across 40 scripted cases, while maintaining 100% correctness. The v2 slot interface increased the full prompt by approximately 59 tokens, but reduced mean observation size from 62.97 to 45.77 tokens per eligible call, a 27.31% reduction. The largest demonstrated financial lever was success rate: GPT-4o Mini v2 improved from 51.79% to 83.93% relative to v1, reducing expected fallback by US\$2.9475 per referral, or US\$11,790 monthly, while Layer 1 increased by only US\$0.000205 per referral.

The archives record provider-side cached tokens for several models, but the headline comparison uses uncached list prices for consistency. The prototype shipped with an eight-turn cap and a 60,000-token per-run ceiling. A monthly per-user limit is 500 referrals every user and production Layer 3 costs remain deployment-policy decisions.

2. Table 1- Main v2 Cost-to-Serve

Model	Passed	Success	Input / Output Tokens	Price in/out per 1M	L1 / Referral	L2 / Referral	Total / Referral	Monthly L1+L2
Gemini 2.5 Flash Lite	55/56	98.2143 %	551,399 / 20,500	\$0.10 / \$0.40	\$0.001131	\$0.163750	\$0.164881	\$659.52
GPT-5 Mini	54/56	96.4286 %	460,490 / 70,676	\$0.25 / \$2.00	\$0.004580	\$0.327500	\$0.332080	\$1,328.32
Qwen 2.5 72B	53/56	94.6429 %	477,867 / 13,829	\$0.36 / \$0.40	\$0.003171	\$0.491250	\$0.494421	\$1,977.68
GPT-4o Mini	47/56	83.9286 %	722,257 / 15,697	\$0.15 / \$0.60	\$0.002103	\$1.473750	\$1.475853	\$5,903.41
Claude 3 Haiku	26/56	46.4286 %	782,724 / 28,958	\$0.25 / \$1.25	\$0.004141	\$4.912500	\$4.916641	\$19,666.56

Monthly total = reported L1+L2 monthly amount + Layer 3 (not measured).

3. Table 2. GPT- 4o Mini v1 vs v2 Controlled Comparison

Metric	v1	v2	Change v2-v1
Trials	56	56	0
Passed	29	47	+18

Success rate	51.7857%	83.9286%	+32.1429pp
Input tokens	650,695	722,257	+71,562
Output tokens	14,433	15,697	+1,264
Battery list-price cost	\$0.106264	\$0.117757	+\$0.011493
L1/referral	\$0.001898	\$0.002103	+\$0.000205
L2/referral	\$4.421250	\$1.473750	-\$2.947500
Total/referral	\$4.423148	\$1.475853	-\$2.947295
Monthly @4,000	\$17,692.59	\$5,903.41	-\$11,789.18

Both versions are recalculated at the same list prices: US\$0.15 input and US\$0.60 output per one million tokens.

4. Table 3. Success-Rate Sensitivity

Model	P – 10 pp: Cost / Monthly	Measured P: Cost / Monthly	P + 10 pp: Cost / Monthly
Gemini 2.5 Flash Lite	88.2143%: \$1.081881 / \$4,327.52	98.2143%: \$0.164881 / \$659.52	100%: \$0.001131 / \$4.52
GPT-5 Mini	86.4286%: \$1.249080 / \$4,996.32	96.4286%: \$0.332080 / \$1,328.32	100%: \$0.004580 / \$18.32
Qwen	84.6429%: \$1.411421 / \$5,645.68	94.6429%: \$0.494421 / \$1,977.68	100%: \$0.003171 / \$12.68
GPT-4o Mini	73.9286%: \$2.392853 / \$9,571.41	83.9286%: \$1.475853 / \$5,903.41	93.9286%: \$0.558853 / \$2,235.41
Claude Haiku	36.4286%: \$5.833641 / \$23,334.56	46.4286%: \$4.916641 / \$19,666.56	56.4286%: \$3.999641 / \$15,998.56

5. Break-even Analysis

Pair selection: Gemini 2.5 Flash Lite is the lowest-L1 model and achieved the lowest measured cost-to-serve; GPT-5 Mini is the next-highest-accuracy reference with a materially higher L1.

$C = \$0.001131069643$; $E = \$0.332079901786$; $F = \$9.17$.

$E - C = \$0.330948832143$.

Break-even failure rate = $(E - C) / F = 3.6090\%$.

Break-even success rate = 96.3910%.

Gemini measured 98.2143%, clearing break-even by 1.8233 percentage points.

6. Table 4. Four Cost Levers

Lever	Before	After	Measured change
Tool block B – D2(a)	1,007 approx. tokens; 6 tools	912 tokens; 5 tools	- 95 tokens (-9.4%)
Turn count T – D2(c)	Serial: 4.625 mean turns; 1,043,400 input tokens	Parallel: 3.625 turns; 749,400 tokens	-21.62% turns; -28.18% input; pass rate unchanged
Observation D – D2(b)	62.971 tokens/call; 3.229 rows	45.771 tokens/call; 1.629 rows	-17.2 tokens/call (-27.31%)
Success rate	v1: 51.7857%; L2 \$4.42125	v2: 83.9286%; L2 \$1.47375	+32.1429 pp; L2 -\$2.9475/referral

7. Table 5. Shipping Caps and Remaining Assumptions

Item	Value	Evidence / status
------	-------	-------------------

Step cap	8 turns/run	Experimental configuration; GPT-4o Mini and Claude each had one 8-turn run. Production validation required.
Per-referral budget ceiling	60,000 total tokens/run	Prototype configuration; observed maximum was 25,627 tokens.
Monthly limit per user	500 referrals/user/month	Deployment policy assumption. At 4,000 referrals/month, this corresponds to eight full-capacity users if workload were evenly distributed. Actual per-user workload was not measured, so this cap should be recalibrated after production usage data are available.
Layer 3	Not measured	Production hosting, storage, monitoring and maintenance costs must be added before deployment.

8. Calculation Audit

Input cost = input tokens / 1,000,000 × input price.

Output cost = output tokens / 1,000,000 × output price.

L1 = battery token cost / trials.

P = passed / trials. L2 = (1 – P) × \$9.17.

Cost-to-serve = L1 + L2.

Monthly L1+L2 = cost-to-serve × 4,000.

Sensitivity replaces P with max(0, P – 0.10) and min(1, P + 0.10).

Break-even success = 1 – (E – C) / F.

Data sources: six D5 model-result JSON files; d2a_tool_block_comparison.json; d2b_comparison_summary.json; d2c_comparison_summary.json;